

AI Practical File questions and instructions

Term - I

1. WAP to accept marks of Artificial Intelligence subject out of 50 from the user for 5 students. Find total, average and percentage from the accepted data. Display total, average and percentage.
2. A cloth showroom announced the following festival discount and assured gifts on the purchase of items based on their total cost.

Total cost	Discount	Assured Gifft
<2000	5%	Wall Clock
2001-3000	10%	School bag
3001-5000	15%	Electric Iron
5000<	20%	Wrist Watch

Accept a value for the total cost of the items purchased from the user. Calculate and display the discount and amount to be paid along with the assured gift.

3. Create a dictionary having the data of any five cities of India and their population. Print the dictionary such that every City with its population should be displayed in a new line.
4. Create arrays as per the given instructions:
 - i. 1D array with any 5 values between 100 to 200.
 - ii. 1D array with any 5 values between 150 to 200.
 - ii. 2D array with the above(i) and (ii) 1D arrays.
5. Create the given dataframe using your choice of method:

Regd. No	Name	Marks%
1000	Steve	86.29
1001	Mathew	91.63
1002	Jose	72.90
1003	Patty	69.23
1004	Vin	88.30

6. A csv file 'cars.csv' contains the following data :

Brand	Model	Year	Price
Toyota	Camry	2021	945000
Honda	Civic	2022	896000
Ford	Mustang	2021	867000
Chevrolet	Malibu	2021	785000
Nissan	Altima	2022	625000

WAP to add more data to the above file.

7.

To write a program to create the given DataFrame and perform the operations listed below:

	Rajat	Dev	Priya	Sam	John
Physics	90	92	89	81	94
Chemistry	91	81	91	71	95
Biology	97	96	88	67	99

- a) To add a new column
- b) To add a New Row
- c) To change values of a row
- d) To delete a column
- e) To delete a row

8. WAP to display the below items from the given dataframe:

	Name	Age	Salary	Post
0	Niti	45	120000	CEO
1	Robin	NaN	73000	Manager
2	Sunita	39	NaN	Developer
3	Shaurya	50	40000	Designer
4	Aarti	60	30000	NaN

- a) Print the number of rows and columns.

- b) Fill in each missing value with the appropriate value of your choice.
- c) Print the first 2 rows.
- d) Print the last 3 rows.
- e) Print the dataframe.

9. Imagine we are collecting library records where some data is missing.

- Ravindra forgot to return the book, so the Return Date is missing.
- Jasmin didn't mention the Book Title while borrowing.
- Kanika didn't update her Membership status as she was on vacation.

Write Python code to:

- a. Check for missing values in the dataset.
- b. Find the total number of missing values.
- c. Drop rows with missing values.
- d. Fill missing values with a default value.

10. WAP to print the following pattern using a nested for loop.

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
*  *  *  *
*  *  *
*  *
*
```

Important Information:

- The Index should be in printed form.
- Outputs should be stuck in portrait form parallel to the practical question.
- The output should have your name, class and section printed as a first line.
- Your practical file must have the following items:
 1. Cover page
 2. Index
 3. Term - I practical questions
- **The submission date on or before: 31/08/2026**